

Homework 19

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19/02/2026

1 *Premises* : $p \implies (q \wedge r), s \implies r, r \implies p$. *Prove* :
 $s \implies q$

$p \implies (q \wedge r)$	<i>Premise</i>	(1)
$s \implies r$	<i>Premise</i>	(2)
$r \implies p$	<i>Premise</i>	(3)
$s \implies p$	2, 3 <i>Hypothetical Syllogism</i>	(4)
$s \implies (q \wedge r)$	4, 1 <i>Hypothetical Syllogism</i>	(5)
$\neg s \vee (q \wedge r)$	5 <i>Material Implication</i>	(6)
$(\neg s \vee q) \wedge (\neg s \vee r)$	6 <i>Distribution</i>	(7)
$\neg s \vee q$	7 <i>Simplification</i>	(8)
$s \implies q$	8 <i>Material Implication</i>	(9)

$\therefore p \implies (q \wedge r), s \implies r, r \implies p$. *Prove* : $s \implies q$

2 *Premises* : $\neg(r \vee s), \neg p \implies s, p \implies q$. *Prove* : q

$$\neg(r \vee s) \quad \text{Premise} \quad (1)$$
$$\neg p \implies s \qquad \text{Premise} \qquad (2)$$
$$p \Rightarrow q \quad \text{Premise} \quad (3)$$
$$\neg s \implies p \qquad 2 \text{ Contrapositive} \qquad (4)$$
$$\neg s \implies q \qquad 3, 4 \text{ Hypothetical Syllogism} \qquad (5)$$
$$\neg r \wedge \neg s \quad 1 \text{ DeMorgan} \quad (6)$$
$$\neg s \qquad \qquad \qquad 6 \textit{ Simplification} \qquad \qquad (7)$$
$$q \qquad 5,7 \text{ ModusPonens} \qquad (8)$$

$\therefore \text{Premises} : \neg(r \vee s), \neg p \Rightarrow s, p \Rightarrow q. \text{Prove} : q$

3 *Premises* : $(p \wedge q), p \implies r, (q \wedge r) \implies s$. *Prove* : s

$p \wedge q$ *Premise* (1)

$p \implies r$ *Premise* (2)

$(q \wedge r) \implies s$ *Premise* (3)

p 1 *Simplication* (4)

r 2, 4 *ModusPonens* (5)

q 1 *Simplication* (6)

$q \wedge r$ 5, 6 *Conjunctive* (7)

s 3, 7 *ModusPonens* (8)

$\therefore$ *Premises* : $(p \wedge q), p \implies r, (q \wedge r) \implies s$. *Prove* : s